

Vishesh Panchal

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Summary

Python Software Engineer with **3+ years** of experience spanning **LLM inference optimization**, distributed GPU systems, and scalable backend engineering. Hands-on expertise in deploying and benchmarking large language models on multi-node GPU clusters (**H200**, **A100**) using **vLLM**, **Ray**, and **SLURM**, with experience in parallelism strategies, **KV-caching**, and inference performance profiling (**TTFT/TPOT**). Proficient in building production-grade backend systems with **FastAPI**, **Flask**, and **AWS**, and integrating GenAI tooling including **LiteLLM**, **RAG**, and **LangChain**.

Work Experience

Software Engineer

Feb 2026 – May 2026

Expedera India Pvt Ltd. – Hyderabad, India

- Developed and optimized LLM inference workloads on a production-grade framework built on **vLLM**, working across **H200 ×8**, **A100 ×4**, and **RTX 4090 ×4** multi-node GPU clusters connected via **Ray**.
- Implemented and evaluated multiple GPU parallelism strategies including **Pipeline**, **Data**, and **Expert Parallelism** for distributed LLM serving across multi-node setups.
- Deployed and managed **Ray** clusters with **SLURM** architecture to efficiently schedule and distribute inference tasks across GPU nodes.
- Built an orchestration framework to benchmark LLM engine configurations covering parallelism strategies, request routing, **KV-caching**, dynamic batching, and **CUDA graph** optimizations.
- Monitored and analyzed inference performance metrics including **TTFT** and **TPOT** to evaluate model serving efficiency across different hardware and configuration setups.

Python Software Engineer

Jun 2025 – Oct 2025

Albiorix Technology – Ahmedabad, India

- Built an **Auth0**-based authentication and **Data Connector** system integrating third-party platforms (**HubSpot**, **Xero**) using **AWS** and **Azure** services, reducing integration setup time significantly.
- Developed the backend for **Fantix**, a real-time sports betting application, using **FastAPI**, **APScheduler**, and **AWS** – supporting live odds streaming, per-match user chat, and a coin-based reward system.

Software Engineer

Apr 2023 – May 2025

Dolcera Pvt. Ltd. – Hyderabad, Telangana

- Developed and deployed the backend of the **IPAuthor OfficeAction** platform using **Python**, **Flask**, and **AWS** (EC2, S3, Lambda, RDS), handling patent drafting workflows at scale.
- Integrated **LiteLLM** and **Retrieval-Augmented Generation (RAG)** with **Qdrant** vector store to improve AI-driven query accuracy for patent document analysis.
- Designed and documented **RESTful APIs**, reducing onboarding time for new developers and improving cross-team communication.
- Implemented **Celery** task queues with **Redis** for asynchronous execution of multi-step AI workflows, improving system throughput.

Django Developer Intern

Dec 2021 – Jun 2022

CreArt Solutions – Hyderabad, Telangana

- Built **DevSpace**, a full-stack idea-sharing platform where users post concepts and businesses can browse and purchase them.
- Implemented user authentication, an admin dashboard, and a feedback system using **Django** and **PostgreSQL**.

Projects

MiniRAG – RAG Experimentation Platform

visheshpanchal/minirag

- Built a full-stack, UI-driven experimentation system for testing and evaluating multiple **RAG** methods, currently supporting naive and hybrid retrieval, with a modular architecture designed for adding new methods incrementally.
- Architected the backend with **LangChain**, **LangGraph**, **Qdrant** for vector storage, and **Celery** with **Redis/PostgreSQL** for asynchronous task processing; containerized the full stack with **Docker** and built the frontend in **React** with **TypeScript**.

Orchestrator – LLM Inference Benchmarking Framework

- Built an internal orchestration framework to benchmark and test **tLLM** (custom **vLLM**-based inference engine) across configurations including **Pipeline**, **Data**, and **Expert Parallelism**, request routing, **KV-caching**, dynamic batching, and **CUDA graph** optimizations on **H200 ×8**, **A100 ×4**, and **RTX 4090 ×4** multi-node GPU clusters.
- Implemented parallel test execution on a single machine using **GPU slicing** with **Docker** containers, with process allocation and scheduling managed via **SLURM**.

IPAuthor – OfficeAction Patent Drafting Platform

- End-to-end AI patent drafting tool using **LiteLLM**, **OpenAI fine-tuning**, **Gemini**, **RAG** (FAISS/Qdrant), and prompt engineering; deployed on **AWS** (EC2, RDS, Lambda, Redis) with a **Flask** API backend.

Fantix Sports – Real-Time Betting Application

- Real-time sports betting backend using **FastAPI** and **DraftKings Odds API**; features include live odds streaming, match-scoped user chat, and a coin-based win/loss ledger; deployed on **AWS**.

Skills

Core Skills: Distributed Systems, LLM Inference Optimization, GPU Parallelism (Pipeline, Data, Expert), Benchmarking & Profiling, Backend Development, Cloud Infrastructure, AI/ML Engineering

Technical Skills: Python, SQL, Flask, FastAPI, Django, Celery, PostgreSQL, MongoDB, Docker, SLURM, Ray, vLLM, LangChain, LangGraph, RAG, FAISS, Qdrant, HuggingFace Transformers, Agentic AI, Graph Engineering, Vector Database, AWS (EC2, S3, Lambda, RDS), Git, GitHub, Machine Learning, Deep Learning

Soft Skills: Analytical Thinking, Problem Solving, Self-Directed Learning, Attention to Detail, Adaptability

Education

Bachelor of Engineering – Computer Science

Aug 2018 – Jun 2022

LJ University – Ahmedabad, Gujarat

- CGPA: 7.76 / 10
- Relevant Coursework: Computer Architecture, Data Structures & Algorithms, Computational Theory, Machine Learning

Master Of Science – Artificial Intelligence and Machine Learning

June 2026 – Expected Nov 2027

Scaler Neovarsity

- Relevant Coursework: SQL, Data Analysis, Machine Learning Algorithm, Deep Learning, LLM, GenAI, Product Analysis, Agents

Achievements

- Solved **200+ problems** on LeetCode, covering arrays, graphs, dynamic programming, and system design concepts.
- Earned **HackerRank Problem Solving Certificate** (Gold badge in Python and Problem Solving).